

ESEMPIO N.2

SUBJECT: QUANTUM CHROMODYNAMICS

TITOLO PAPER: "Non-monotonic energy dependence of net-proton number fluctuations"

query con cui ho interrogato inspire hep:

<https://inspirehep.net/api/literature?sort=mostcited&size=25&page=1&q=t%20quantum%20chromodynamics%20and%20date%202016-%3E2026>

OUTPUT INSPIRE:

(Considero solo la entry del paper preso in considerazione), il paper scelto ha poche figure apposta per non sprecare api call.

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-----ANALISI PAPER SENZA PIPELINE VISIVA-----

ANALISI PDF:

comando: python main.py analyze --pdf downloaded_pdfs/"1850675_Nonmonotonic Energy Dependence of Net-Proton Numbe.pdf" --model gemini

OUTPUT:

Output JSON:

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{
  "problem_addressed": "Search for the QCD critical point by examining non-monotonic behavior of higher-order moments (specifically kurtosis times variance) of net-proton number fluctuations as a function of collision energy in heavy-ion collisions.",
  "main_contributions": [
    "First evidence of non-monotonic variation in  $\kappa \sigma^2$  of net-proton distribution with 3.1\% significance for central Au+Au collisions.",
    "Measurement of net-proton cumulants and ratios across a wide range of  $\sqrt{s_{NN}}$  from 7.7 to 200 GeV using STAR detector at RHIC (BES phase I).",
    "Demonstration that peripheral collisions and theoretical models without a critical point (HRG, UrQMD) show monotonic energy dependence, supporting the critical point interpretation."
  ],
  "methodology": "Au+Au collisions at various  $\sqrt{s_{NN}}$  were recorded with the STAR detector. Protons and anti-protons were identified via TPC and TOF within  $|y| < 0.5$  and  $0.4 < p_T < 2.0$  GeV/c. Event centrality was determined from charged particle multiplicity. Net-proton cumulants up to fourth order were calculated, corrected for volume fluctuations and detector efficiency (binomial and unfolding methods). Ratios  $S = C_3/C_2$  and  $\kappa \sigma^2 = C_4/C_2$  were obtained as functions of  $\sqrt{s_{NN}}$ . Fourth- and fifth-order polynomial fits were applied to  $\kappa \sigma^2$  and  $S$  data (excluding 200 GeV point for  $\kappa \sigma^2$  fit). The significance of non-monotonic variation was assessed via the derivative of the polynomial fit using bootstrap sampling of statistical and systematic uncertainties.",
  "datasets_used": [
    "STAR Beam Energy Scan Phase I Au+Au collisions at  $\sqrt{s_{NN}} = 7.7, 11.5, 14.5, 19.6, 27, 39, 54.4, 62.4$ , and 200 GeV (2010-2017).",
    "Peripheral Au+Au collisions (70-80% centrality) at the same energies for comparison.",
    "Reference data from HADES experiment for 0-10% Au+Au collisions at  $\sqrt{s_{NN}} = 2.4$  GeV (shown in Fig.4)."
  ],
  "experiments": "The study measured event-by-event net-proton distributions and computed cumulants  $C_1$ - $C_4$  for central (0-5%) and peripheral (70-80%) Au+Au collisions across nine collision energies. The cumulants were corrected for volume fluctuations and detector inefficiencies. The ratios  $S$  and  $\kappa \sigma^2$  were plotted versus  $\sqrt{s_{NN}}$  and fitted with polynomial functions. The derivative of the polynomial fit for  $\kappa \sigma^2$  was evaluated to quantify non-monotonic behavior, with significance determined by varying data points within uncertainties and counting pseudo-experiments where derivative sign changed. Model comparisons (HRG, UrQMD, JAM) were performed using  $\chi^2$  tests."
```

"results": [

"Observed non-monotonic variation in $\langle S \rangle$ for central 0-5% Au+Au collisions with a significance of 3.1 relative to Skellam expectation.",

"The derivative of the fourth-order polynomial fit to $\langle S \rangle$ changes sign between $\sqrt{s_{NN}}$ 19.6 and 27 GeV, indicating a minimum in the kurtosis-times-variance.",

"Peripheral collisions and model calculations (HRG GCE, HRG CE, HRG EV, UrQMD, JAM) show monotonic energy dependence and do not exhibit the non-monotonic trend.",

"C1 and C3 decrease monotonically with increasing $\sqrt{s_{NN}}$ for central collisions.",

"Within 1.0 significance, S allows for a non-monotonic energy dependence.",

"The χ^2 test p-values between data and models (for $\sqrt{s_{NN}}$ =7.7-27 GeV) are: HRG GCE <0.001 for both S and $\langle S \rangle$; HRG EV <0.001 for S and 0.0145 for $\langle S \rangle$; HRG CE 0.0754 for S and 0.0450 for $\langle S \rangle$; UrQMD <0.001 for both."

],

"limitations": [

"Systematic uncertainties from proton and anti-proton reconstruction efficiency, particularly the different efficiency at $\sqrt{s_{NN}}$ =54.4 GeV leading to deviation from general trend.",

"The $\sqrt{s_{NN}} = 200$ GeV data point was not included in the polynomial fit for $\langle S \rangle$ due to non-convergence or larger χ^2 values, limiting the high-energy leverage.",

"Dependence on model assumptions for baseline comparisons (e.g., HRG, UrQMD).",

"Finite statistics at lower energies restrict precision."

],

"future_work": [

"Higher event statistics from Beam Energy Scan Phase II (BES-II) will enable more differential measurements in rapidity and transverse momentum (y - p_T) and improve comparisons with dynamical QCD calculations.",

"Reduced uncertainties will allow a more precise determination of the significance of the non-monotonic signal and help establish the QCD critical point."

],

"relevant_elements": [

"Cumulant definitions: $C_1 = M$, $C_2 = \langle S^2 \rangle$, $C_3 = \langle S^3 \rangle$, $C_4 = \langle S^4 \rangle$ ",

"Ratios: $S = C_3/C_2$, $\langle S \rangle = C_4/C_2$ ",

"Fourth order polynomial fit for $\langle S \rangle$ versus $\sqrt{s_{NN}}$ ",

"Fifth order polynomial fit for S versus $\sqrt{s_{NN}}$ ",

"Table I: χ^2 p-values between data and various models for $\sqrt{s_{NN}}$ =7.7-27 GeV",

"Table II: Parameters of fourth/fifth order polynomial describing collision energy dependence of $\langle S \rangle$ and S ",

"Table III: Values of derivative of polynomial for $\langle S \rangle$ and S at various $\sqrt{s_{NN}}$ ",

"Table IV: Values of derivative of fourth order polynomial describing $\langle S \rangle$ versus $M/\sqrt{s_{NN}}$ ",

"Table V: Same as Table III with uniform distribution assumption for uncorrelated systematic uncertainties",

"Fig.1: Event-by-event net-proton number distributions for nine $\sqrt{s_{NN}}$ values",

"Fig.2: Net-proton cumulants (C_n) for central and peripheral collisions",

"Fig.3: S and $\langle S \rangle$ with polynomial fits and their derivatives",

"Fig.4: S and $\langle S \rangle$ compared to HRG, UrQMD, JAM models and HADES data",

"Fig.5: Particle identification via TOF and TPC",

"Fig.6: Efficiency-corrected cumulants using unfolding method",

"Fig.7: Magnified peripheral collision cumulants",

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    "Fig.8: Rapidity dependence of cumulant ratio  $C_4/C_2$ ",
    "Fig.9: Significance of deviation in  $\langle S \rangle$  values from models and peripheral collisions",
    "Skellam baseline expectation",
    "Hadron Resonance Gas (HRG) model (GCE, EV, CE variants)",
    "UrQMD transport model",
    "JAM microscopic transport model"
  ],
  "visual_analysis_findings": []
}

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ANALISI LATEX:

comando : python main.py analyze --pdf downloaded_pdfs/"1850675_Nonmonotonic Energy Dependence of Net-Proton Number" --model gemini

Output JSON:

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{
  "problem_addressed": "Search for signatures of the QCD critical point by examining non-monotonic variations with collision energy of higher order moments (skewness and kurtosis) of net-baryon number distributions, using net-proton fluctuations as a proxy.",
  "main_contributions": [
    "First evidence of a non-monotonic variation of  $\langle S \rangle$  (kurtosis times variance) with  $\sqrt{s_{NN}}$  in central Au+Au collisions at 3.1 GeV significance.",
    "Demonstration that peripheral collisions and non-critical theoretical baselines (HRG models, UrQMD) do not exhibit the observed non-monotonic trend.",
    "Comprehensive measurements of net-proton cumulants up to the fourth order, including corrections for volume fluctuations, detector efficiencies, and centrality bin width effects, extending the STAR Beam Energy Scan data set."
  ],
  "methodology": "STAR detector measured protons and antiprotons in the kinematic range  $0.4 < p_T < 2.0$  GeV/c and  $|y| < 0.5$  from Au+Au collisions at nine  $\sqrt{s_{NN}}$  energies (7.7-200 GeV) during the RHIC BES Phase I (2010-2017). Central (0-15%) and peripheral (70-180%) events were selected based on charged-particle multiplicity excluding protons and antiprotons. Particle identification used TPC dE/dx and TOF. Cumulants  $C_n$  were computed and corrected for finite impact parameter volume fluctuations (using multiplicity-dependent corrections) and for detector efficiencies assuming a binomial response. The ratios  $S = C_3/C_2$  and  $\langle S \rangle = C_4/C_2$  were used as observables.",
  "datasets_used": [
    "Au+Au collisions at  $\sqrt{s_{NN}} = 7.7, 11.5, 14.5, 19.6, 27, 39, 54.4, 62.4, 200$  GeV (STAR BES Phase I)",
    "Central (0-15%) and peripheral (70-180%) event classes",
    "Data taking period 2010-2017 (2483 million events at 7.7 GeV up to 2485 million events at 54.4 GeV)"
  ],
  "experiments": "RHIC Beam Energy Scan (BES) program, Phase I, using the STAR time-of-flight detector at the Relativistic Heavy Ion Collider.",
  "results": "Non-monotonic  $\langle S \rangle$  vs  $\sqrt{s_{NN}}$  in central collisions (significant 3.1% change of the fitted polynomial derivative).  $S$  shows marginal non-

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monotonicity. Peripheral collisions remain monotonic and below the statistical baseline of unity. Comparisons with HRG (GCE, EV, CE) and UrQMD give low χ^2 test p-values ($\chi^2/20.05$), indicating poor agreement. Central collisions exhibit large non-Gaussian cumulants C_3 and C_4 , consistent with expectations near a QCD critical point.",

"limitations": "Only net-proton (a proxy for net-baryon) measured within limited acceptance ($0.4 < p_T < 2.0$ GeV/c, $|y| < 0.5$). Efficiency corrections rely on a binomial response model; residual detector effects may remain. Finite system size and evolution time smear critical signatures. Peripheral collisions may not probe the relevant region of the QCD phase diagram. Statistical and systematic uncertainties still limit precision; full event-by-event net-baryon measurements are needed.",

"future_work": "Collect higher statistics in BES-II and future runs to perform differential studies in rapidity and transverse momentum, enabling more precise comparisons with QCD-based calculations that incorporate dynamical evolution, and to firmly establish the presence of a critical point.",

"relevant_elements": [

"Cumulant definitions: $C_1 = \langle N \rangle$, $C_2 = \langle N^2 \rangle - \langle N \rangle^2$, $C_3 = \langle N^3 \rangle - 3\langle N \rangle \langle N^2 \rangle + 2\langle N \rangle^3$, $C_4 = \langle N^4 \rangle - 4\langle N \rangle \langle N^3 \rangle + 6\langle N \rangle^2 \langle N^2 \rangle - 3\langle N \rangle^4$ ",

"Skewness and kurtosis: $S = \langle N^3 \rangle / \langle N \rangle^3$, $\kappa = \langle N^4 \rangle / \langle N \rangle^4$ ",

"Ratio observables: $S^2 = C_3^2 / C_2^3$, $\kappa - 3 = C_4 / C_2^2$ ",

"Binomial efficiency correction for detector response",

"Fifth-order polynomial fit to S^2 and fourth-order polynomial fit to $\kappa - 3$ ",

"Table 1: χ^2 test p-values for S^2 and $\kappa - 3$ between data and HRG GCE, HRG EV, HRG CE, UrQMD"

],

"visual_analysis_findings": []

}

ANALISI CON PIPELINE VISIVA

- PDF

comando: python main.py full-analyze --pdf downloaded_pdfs/"1850675_Nonmonotonic Energy Dependence of Net-Proton Numbe.pdf" --model gemini

Output JSON:

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{
  "problem_addressed": "Searching for the Quantum Chromodynamics (QCD) critical point by studying non-monotonic variations in higher moments of net-proton number distributions as a function of collision energy ( $\sqrt{s_{NN}}$ ) in heavy-ion collisions at RHIC.",
  "main_contributions": [
    "First evidence of non-monotonic variation in kurtosis times variance ( $\kappa - 3$ ) of net-proton distribution with 3.1% significance",
    "Improved proton identification with  $p_T$  acceptance up to 2 GeV/c using TPC and TOF detectors",
    "Cumulant measurements up to fourth order for net-proton distributions at nine collision energies"
  ],
```

"methodology": "Measured Au+Au collisions using STAR detector at RHIC over $\sqrt{s_{NN}} = 7.7$ -200 GeV. Classified events by centrality (0-5% central, 70-80% peripheral). Identified protons/anti-protons via TPC dE/dx (up to 800 MeV/c) and TOF (up to 2 GeV/c). Calculated cumulants C1-C4 from event-by-event net-proton distributions in $|y| < 0.5$, $0.4 < p_T < 2.0$ GeV/c. Applied corrections for efficiency, volume fluctuations, and centrality bin width. Used binomial detector response model with unfolding validation.",

"datasets_used": [

"Au+Au collisions at $\sqrt{s_{NN}} = 7.7, 11.5, 14.5, 19.6, 27, 39, 54.4, 62.4, 200$ GeV",

"STAR Beam Energy Scan (BES) phase-I data from 2010-2017",

"Central collisions (0-5% based on multiplicity)",

"Peripheral collisions (70-80% based on multiplicity)"

],

"experiments": "Statistical uncertainties via bootstrap and Delta theorem methods. Systematic uncertainties from varying track reconstruction requirements (DCA, space points, PID criteria). Cross-checked with HRG models (GCE, CE, EV), UrQMD transport model, and JAM model. Used χ^2 tests comparing data to models.",

"results": "Observed non-monotonic variation of S/C_2 with $\sqrt{s_{NN}}$ in central Au+Au collisions with 3.1% significance. S/C_2 deviates from statistical baseline of unity with sign change in derivative indicating peak structure. Peripheral collisions show monotonic decrease of S/C_2 below unity. Other moment products ($S/C_2 = C_3/C_2$, $S/C_2/M$) show weaker non-monotonic behavior. Models without critical point (HRG, UrQMD) cannot explain the observed energy dependence.",

"limitations": "Significance from Skellam expectation (not Poisson baseline). Statistical hadron gas and transport models show better agreement for lower collision energies but cannot explain high-energy data. Finite size and time effects may obscure critical signatures. Limited by detector acceptance and efficiency corrections.",

"future_work": "Higher event statistics for differential measurements in y-pT space. Improved comparison with QCD calculations incorporating heavy-ion dynamics. Extension of BES program with larger data samples.",

"relevant_elements": [

"Definition: $S/C_2 = C_3/C_2 = (\langle n^3 \rangle / \langle n \rangle^3) / (\langle n^2 \rangle / \langle n \rangle^2)$ ",

"Definition: $S/C_2 = C_4/C_2 = (\langle n^4 \rangle / \langle n \rangle^4) / (\langle n^2 \rangle / \langle n \rangle^2)$ ",

"Figure 1: Net-proton number distributions vs $\sqrt{s_{NN}}$ ",

"Figure 2: Cumulants C1-C4 vs collision energy",

"Figure 3: S/C_2 and $S/C_2/M$ with polynomial fits and derivatives",

"Figure 4: Comparison with HRG and UrQMD models",

"Table I: χ^2 p-values for model comparisons",

"Table II: Polynomial fit parameters for S/C_2 and $S/C_2/M$ ",

"Table III: Derivative values and significances",

"Supplemental Material figures 6-9",

"Eq. for net-proton deviation: $\langle n^3 \rangle - 3\langle n \rangle \langle n^2 \rangle$ "

],

"visual_analysis_findings": [

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"AI curve-fit approximation of Figure 1, curve '62.4 (red stars)': $y = 0.18811703/(x_0 + \exp(1.2509168 - x_0))$ ($R^2=0.673$, low confidence)",

```

    "AI curve-fit approximation of Figure 1, curve '54.4 (red pluses)': y =
-0.05967525/(x0*1.3744402*(x0*x0*0.0063782623*(-0.0634772)*x0 + 0.24995252) - 2.1337984) (R\
u00b2=0.878)",
    "AI curve-fit approximation of Figure 1, curve '39 (blue triangles)': y =
3.4408224/(9.892098*(exp(x0/4.323362) + 1.9738944/exp(x0 - 1*5.1972394))) (R\u00b2=0.973)",
    "AI curve-fit approximation of Figure 1, curve '27 (blue squares)': y = -0.00029782296*x0*x0 +
x0*0.0064917733 (R\u00b2=0.553, low confidence)",
    "AI curve-fit approximation of Figure 1, curve '19.6 (blue circles)': y = exp(-0.2008174*x0 -
2.4372568/(x0*(x0*0.015513207)*(x0*x0*0.015513207)*(x0*x0*0.018857222)*(x0*0.043298844 -
1*0.06443104))) (R\u00b2=0.922)",
    "AI curve-fit approximation of Figure 1, curve '14.5 (black stars)': y = 0.7923345*x0 +
(x0*(0.00058207626*x0 - 0.199398335) - 1.59497469)*log(x0) + 0.2176881 (R\u00b2=0.806)",
    "AI curve-fit approximation of Figure 1, curve '11.5 (black squares)': y = x0*x0*(-
0.04108372)*(0.21096818 - 0.01827313*x0)*(0.33540046 - 0.009439284*x0)*(0.33540076 -
0.010765301*x0) (R\u00b2=0.840)",
    "AI curve-fit approximation of Figure 1, curve 'lowest energy (black circles)': y = (log(x0) -
6.385939 + 9.994632/log(x0))*log(12.486389/x0) (R\u00b2=0.701)",
    "AI curve-fit approximation of Figure on page 5, curve 'S\u03c3 polynomial fit': y = 0.12724255 +
19.066626/(x0 + x0 + 17.501148/(x0*((x0 + x0)/69.28346)))) (R\u00b2=0.995)",
    "AI curve-fit approximation of Figure on page 5, curve '\u03ba\u03c3\u00b2 polynomial fit': y = -
x0*(-(-0.00076362735)*x0 - 0.09405257 + 13.9641075/x0) + 11.30351 + 31.510712/x0 (R\
u00b2=0.980)",
    "AI curve-fit approximation of Figure on page 8, discrete data points extracted for C4/C2 vs
rapidity acceptance at various energies with (rapidity, value) coordinates for 7.7, 11.5, 14.5, 19.6, 27,
39, 54.4, 62.4, and 200 GeV"
]
}

```

LATEX:

comando: python main.py full-analyze --latex downloaded_latex/"1850675_Nonmonotonic Energy Dependence of Net-Proton Numbe_source.tar.gz" --model gemini

----- Gemini -----

Model used: nex-agi/nex-n2.5-mini:free

Output JSON:

```

{
  "problem_addressed": "The paper addresses the search for the QCD critical point by examining
whether net-baryon fluctuation observables, measured with net-protons as a proxy, vary non-
monotonically with heavy-ion collision energy across the QCD phase diagram.",
  "main_contributions": [
    "Reported first evidence for a non-monotonic energy dependence of the net-proton kurtosis-times-
variance observable, kappa sigma squared, in central Au+Au collisions at RHIC.",
    "Measured net-proton cumulants and cumulant ratios over a wide range of beam energies and
centrality classes using the STAR detector.",
    "Compared central-collision results with peripheral collisions and with hadron-resonance-gas and
transport-model baselines that do not include a critical point."
  ],

```

"methodology": "The study used STAR measurements of Au+Au collisions from 2010 to 2017 at $\sqrt{s_{NN}} = 7.7, 11.5, 14.5, 19.6, 27, 39, 54.4, 62.4,$ and 200 GeV. Protons and antiprotons were reconstructed in the TPC and identified using dE/dx and TOF information in the acceptance $|y| < 0.5$ and $0.4 < p_T < 2$ GeV/c. Events were classified as 0-5% central or 70-80% peripheral using charged-particle multiplicity, with proton and antiproton candidates excluded from the centrality variable to reduce self-correlations. Cumulants of the event-by-event net-proton number were calculated, corrected for volume fluctuations and finite detector efficiencies, and analyzed through the ratios $C_3/C_2 = S$ sigma and $C_4/C_2 = \kappa$ sigma squared. The energy dependence was fitted with fourth- and fifth-order polynomials, and the sign changes of their derivatives were tested using statistical and systematic uncertainty propagation.",

"datasets_used": [

"STAR Beam Energy Scan Phase-I Au+Au collision data from 2010 to 2017 at $\sqrt{s_{NN}} = 7.7, 11.5, 14.5, 19.6, 27, 39, 54.4, 62.4,$ and 200 GeV.",

"HADES Au+Au data at $\sqrt{s_{NN}} = 2.4$ GeV, shown as a comparison point."

],

"experiments": "The authors measured event-by-event net-proton distributions and cumulants for central and peripheral Au+Au collisions, corrected for multiplicity and detector-efficiency effects, and compared the resulting S sigma and κ sigma squared values with Poisson or Skellam baselines, HRG model calculations, UrQMD, and JAM. The non-monotonicity was quantified by fitting the central-collision data from 7.7 to 62.4 GeV and testing derivative sign changes through random sampling of the data uncertainties.",

"results": "The central 0-5% Au+Au data show a non-monotonic dependence of κ sigma squared on $\sqrt{s_{NN}}$ with a reported significance of 3.1 sigma. The peripheral 70-80% data do not show a non-monotonic trend and remain below the statistical baseline of unity. S sigma allows a non-monotonic dependence only within about 1 sigma, while sigma squared/ M strongly favors monotonic behavior and excludes a non-monotonic trend at 3.4 sigma. Central-collision C_3 and C_4 values are large and indicate non-Gaussian distributions, whereas peripheral values are small and close to zero. HRG and transport models without a critical point generally predict monotonic behavior and do not simultaneously describe the measured central-collision S sigma and κ sigma squared dependences. For 7.7 to 27 GeV, the chi-square p values for the model comparisons are: S sigma = less than 0.001 for HRG GCE, HRG EV, and UrQMD and 0.0754 for HRG CE; κ sigma squared = 0.00553 for HRG GCE, 0.0145 for HRG EV, 0.0450 for HRG CE, and 0.0221 for UrQMD.",

"limitations": "The authors note that finite system size and lifetime can suppress critical-point signatures, that the location of the QCD critical point remains theoretically uncertain, and that lattice QCD calculations face difficulties at finite baryon chemical potential. The 200 GeV point was excluded from the polynomial non-monotonicity analysis because the fits did not converge or gave larger chi-square values. The models without a critical point do not simultaneously explain both measured cumulant ratios, and the observations are described as qualitatively consistent with, rather than a definitive establishment of, a critical point.",

"future_work": "Higher event statistics are needed for more differential measurements in rapidity and transverse momentum and for improved comparisons with QCD calculations that include heavy-ion collision dynamics.",

"relevant_elements": [

"Skewness and kurtosis definitions: $S = \langle(\delta N)^3\rangle/\sigma^3$ and $\kappa = \langle(\delta N)^4\rangle/\sigma^4 - 3$.",

"Cumulant definitions: $C_1 = M$, $C_2 = \sigma^2$, $C_3 = S \sigma^3$, and $C_4 = \kappa \sigma^4$.",

"Baryon-number susceptibility definition: $\chi_n^B = d^n P / d \mu_B^n$.",

"Cumulant-ratio observables: $S \sigma = C_3/C_2 = (\chi_3^B/T)/(\chi_2^B/T^2)$ and $\kappa \sigma^2 = C_4/C_2 = \chi_4^B/(\chi_2^B/T^2)$.",

"Polynomial analysis: $\kappa \sigma^2$ was fitted with a fourth-order polynomial and $S \sigma$ with a fifth-order polynomial over $\sqrt{s_{NN}} = 7.7$ to 62.4 GeV; derivative sign changes were used to test non-monotonicity.",

"Table of chi-square p values for 0-5% central Au+Au collisions over 7.7 to 27 GeV, comparing $S \sigma$ and $\kappa \sigma^2$ with HRG GCE, HRG EV, HRG CE, and UrQMD.",

"Supplemental tables of polynomial coefficients, derivative values, and robustness checks using third-order fits and M/σ^2 as the energy proxy."

],

"visual_analysis_findings": [

"External AI-generated curve-fit approximation of Figure 1, curve 7.7: $y = x_0 \cdot (0.00073131255 - 4.437965e-7 \cdot x_0^{**2}) \cdot \log(x_0)$, R^2 fit=0.328, holdout=0.500. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 11.5: $y = x_0 \cdot (-0.29770064) \cdot (-0.12163098) / ((-x_0 / ((-15.833286) \cdot 0.2084719) - (-9.77664 - 7.2540655 \cdot (-15.805457/x_0))) / 0.19853038 - 9.778422 \cdot x_0)$, R^2 fit=0.767, holdout=0.845. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 14.5: $y = x_0 \cdot (0.005517102 + x_0 \cdot (-0.00018869409))$, R^2 fit=0.262, holdout=0.334. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 19.6: $y = (x_0 \cdot (-0.0026436658) - 1 \cdot (-0.018012552)) \cdot \exp(x_0) \cdot (-0.0076262876) \cdot \exp(x_0 \cdot (-0.03921138) \cdot (x_0 - 0.06860041)) - 0.0006036707$, R^2 fit=0.979, holdout=0.958. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 27: $y = (x_0 + x_0 - 1 \cdot 6.9669764e-10) \cdot \exp(x_0 + x_0 / ((-18.486687/x_0))) \cdot 4.9603674e-8 \cdot \exp(x_0 + x_0 / ((-28.974104/x_0)))$, R^2 fit=0.988, holdout=0.987. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 39: $y = -0.8538367 / (x_0 \cdot (x_0 - 1 \cdot 119.37288) / \exp(x_0 / 1.8343284) + 1.6825315 / ((-226.9797 \cdot \exp(-0.61828492337564 \cdot x_0))))$, R^2 fit=0.980, holdout=0.995. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 54.4: $y = x_0 / (x_0 / (x_0 \cdot (x_0 \cdot 0.00139169 - 1 \cdot (-0.005512774)) - 1 \cdot 0.0049465955) - (x_0 - 0.008408377 \cdot \exp(x_0)))$, R^2 fit=0.981, holdout=0.964. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 62.4: $y = x_0 \cdot (x_0 / (-2.6412637) - 2.6412637) / (\exp(x_0) \cdot (-0.13895711) - 253.42934)$, R^2 fit=0.958, holdout=0.968. This formula was not derived or stated by the paper's authors.",

"External AI-generated curve-fit approximation of Figure 1, curve 200: $y = (x_0 - 1 \cdot 1.724546) \cdot 0.049592853 / ((0.4586799 - \exp(x_0 \cdot (-1.2936149)) / 0.23732002) \cdot \exp(x_0 \cdot 0.06381172 \cdot x_0))$, R^2 fit=0.991, holdout=0.989. This formula was not derived or stated by the paper's authors."

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